

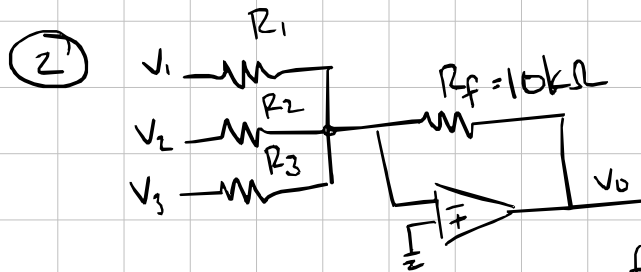
① - 60SD only: AD8613 Parameters

parameters from datasheet:

$$GBP = f_t = 400 \text{ kHz} \quad (\text{Page 3})$$

$$e_{nw} \approx 20 \text{ nV}/\sqrt{\text{Hz}}, \quad f_{le} \approx 100 \text{ Hz} \quad (\text{from fig 25})$$

$$i_{nw} = 0.05 \text{ pA}/\sqrt{\text{Hz}} \quad (\text{page 3})$$



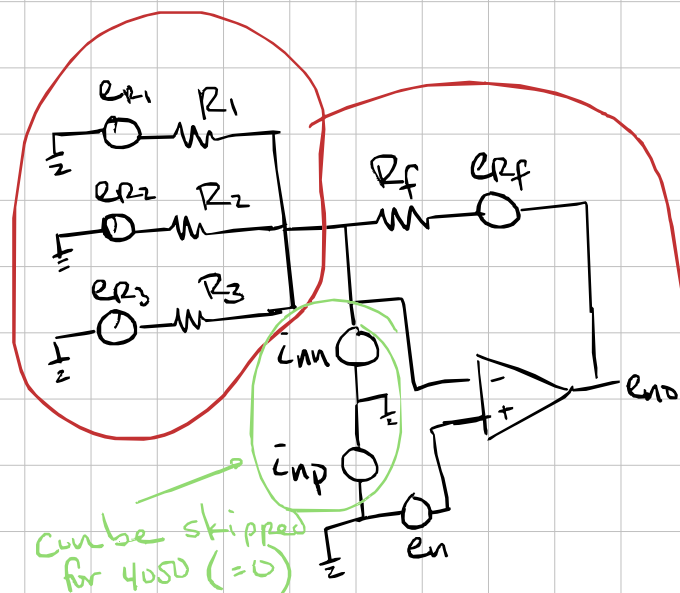
$$V_o = -(V_1 + V_2/2 + V_3/3)$$

from summer equation:

$$R_1 = R_f = 10k\Omega$$

$$R_2 = 2R_f = 20k\Omega$$

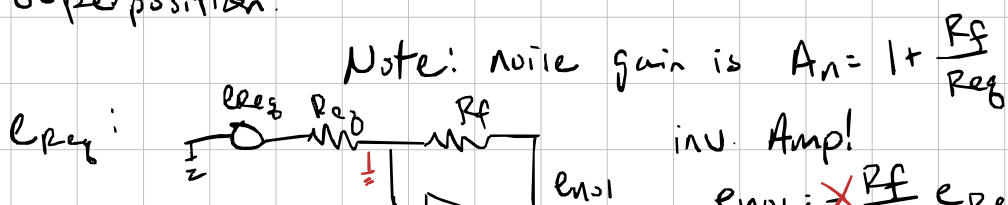
$$R_3 = 3R_f = 30k\Omega$$



R_1, R_2, R_3 in parallel,
 $\therefore R_{eq} = 5.45\Omega$

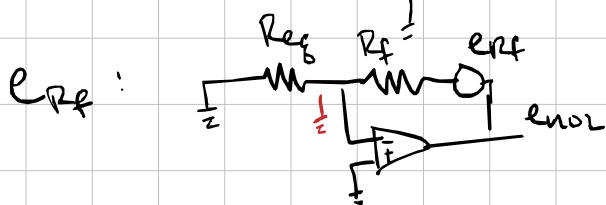
$$e_{R_{eq}}^2 = 4kT R_{eq}$$

Method 1: find e_{no} through direct means:
 Superposition!

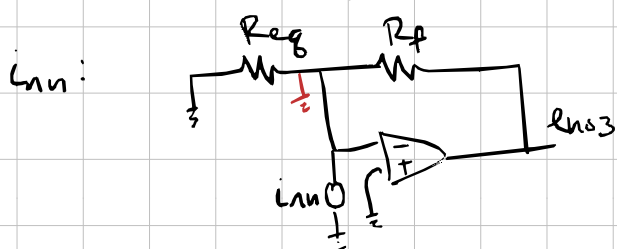


$$e_{no1} = \cancel{\frac{R_f}{R_{eq}}} e_{R_{eq}}$$

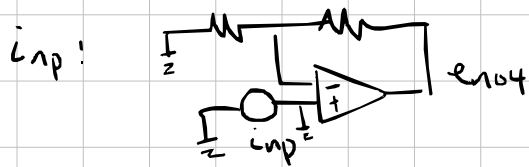
noise doesn't care



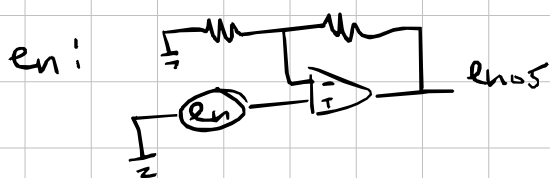
No current in R_{eq}, R_f
 $e_{no2} = e_{R_f}$



No current through R_{eq} :
 $0 = i_{nn} R_f$
 $e_{no3} = i_{nn} R_f$



i_{np} grounded on both sides!
 $e_{no4} = 0$



Non-Inv. Amp!
 $e_{no5} = \left(1 + \frac{R_f}{R_{eq}}\right) e_n = A_n e_n$

$$e_{no}^2 = A_n^2 e_n^2 + \frac{R_f^2}{R_{eq}^2} e_{R_{eq}}^2 + e_{R_f}^2 + i_{nn}^2 R_f^2$$

$$e_{no} = A_n e_{ni} \rightarrow e_{ni} = e_n^2 + \left(\frac{1}{1 + R_f/R_{eq}}\right)^2 \left[\frac{R_f^2}{R_{eq}^2} e_{R_{eq}}^2 + e_{R_f}^2 + i_{nn}^2 R_f^2 \right]$$

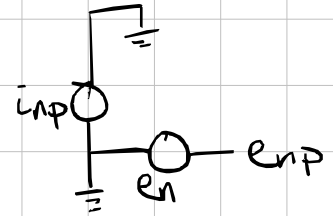
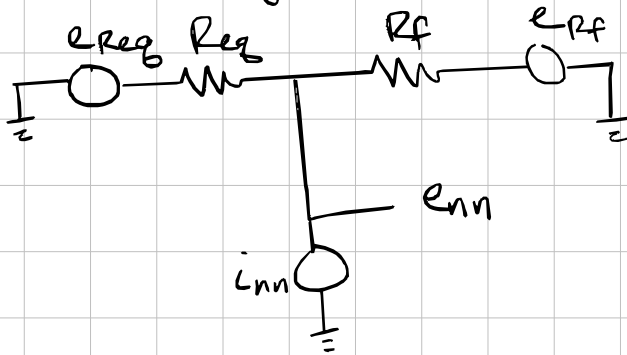
$$e_{ni} = e_n^2 + \left(\frac{R_{eq}}{R_{eq} + R_f}\right)^2 \left[\frac{R_f^2}{R_{eq}^2} 4kT \cancel{R_{eq}} + 4kT R_f^2 + i_{nn}^2 R_f^2 \right]$$

$\uparrow R_{eq}/R_{eq} + R_f$

$$e_{ni} = e_n^2 + \left(\frac{R_{eq} R_f}{R_{eq} + R_f}\right) [8kT + i_{nn}^2]$$

② cont.

Method 2: by eni shortcut:



$$e_{nn}^2 = e_{Reg}^2 \left(\frac{R_F}{R_{Reg} + R_F} \right)^2 + e_{Rf}^2 \left(\frac{R_{Reg}}{R_{Reg} + R_F} \right)^2 + i_{nn}^2 \left(\frac{R_{Reg} R_F}{R_{Reg} + R_F} \right)^2$$

$\uparrow$ V divider $\uparrow$ V divider $\uparrow$ $R_{Reg} \parallel R_F$

$e_{np}^2 = e_n^2$ (inp grounded on each side)

$$e_{ni}^2 = e_n^2 + \left(\frac{1}{R_{Reg} + R_F} \right)^2 \left[e_{Reg}^2 R_F^2 + e_{Rf}^2 R_{Reg}^2 + i_{nn}^2 R_{Reg}^2 R_F^2 \right]$$

$$= e_n^2 + \left(\frac{1}{R_{Reg} + R_F} \right)^2 \left[4kT R_{Reg}^2 R_F^2 + 4kT R_{Reg}^2 R_F^2 + i_{nn}^2 R_{Reg}^2 R_F^2 \right]$$

$$e_{ni}^2 = e_n^2 + \left(\frac{R_{Reg} R_F}{R_{Reg} + R_F} \right)^2 \left[8kT + i_{nn}^2 \right]$$

→

with "ideal" e_{ni} (ideal amplifier considered) we can add in amplifier effects:

$$e_{no} = |A_n(f)| e_{ni} \quad \beta = \frac{R_{eq}}{R_{eq} + R_f}$$

$$f_B / \beta \approx f_t \rightarrow f_B = \beta f_t = f_t R_{eq} / (R_{eq} + R_f)$$

$$A_n = 1 + R_f / R_{eq} / (1 + jf/f_B) \quad |A_n| = \frac{1 + R_f / R_{eq}}{\sqrt{1 + f^2 / f_B^2}}$$

for both 4050/6050: $\beta = 0.3528$, $A_n = 2.83 \text{ V/V}$

$$e_{ni}^2 = e_n^2 + \left(\frac{R_f R_{eq}}{R_{eq} + R_f} \right)^2 [4kT + i_{nn}^2]$$

$$= e_n^2 + (3.53 \text{ k}\Omega)^2 [1.645 \times 10^{-20} \text{ A}^2 + i_{nn}^2]$$

4050 GBP = $f_t = 3 \text{ MHz} \rightarrow f_B \approx 1.06 \text{ MHz}$

$$e_n^2 = (20 \text{ nV}/\sqrt{\text{Hz}})^2 \left(\frac{1 \text{ kHz}}{f} + 1 \right)$$

$i_{nn} = i_{np} = 0$ ← not technically required if $i_{nw} = 0$

$$e_{ni}^2 = \frac{4 \times 10^{-13} \text{ V}^2}{f} + \underbrace{\left(4 \times 10^{-16} \frac{\text{V}^2}{\text{Hz}} + 2.05 \times 10^{-13} \frac{\text{V}^2}{\text{Hz}} \right)}_{\approx 2.5 \times 10^{-12} \frac{\text{V}^2}{\text{Hz}}}$$

$$e_{no}^2 = |A_n(f)|^2 e_{ni}^2$$

$$e_{no}^2 = \frac{2.83^2 \text{ V}^2}{1 + f^2 / 1.06^2} \left(\frac{4 \times 10^{-13} \text{ V}^2}{f} + 2.05 \times 10^{-13} \frac{\text{V}^2}{\text{Hz}} \right)$$

$$E_{no}^2 = \int_{10 \text{ Hz}}^{100 \text{ MHz}} e_{no}^2(f) df$$

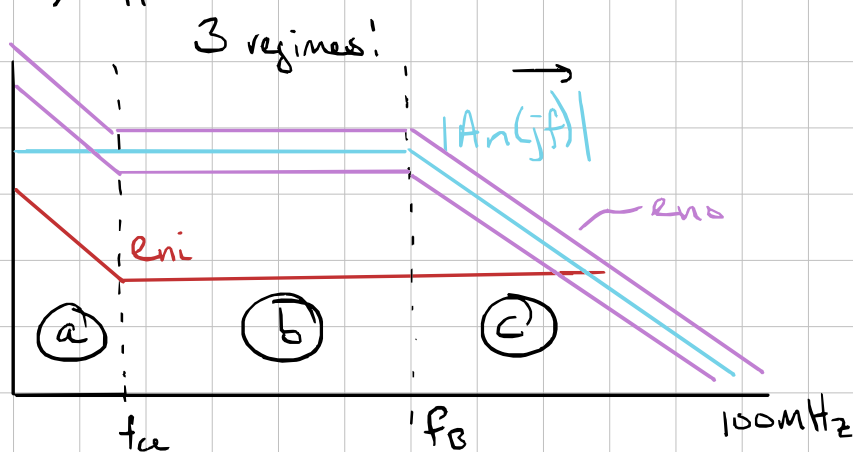
solve 1) Analytically! No thanks

2) MATLAB numerical integration:

$$E_{no}^2 = 2.72 \times 10^{-6} \text{ V}^2$$

$$E_{no} = 1.65 \text{ mV}$$

3) Approximate!



Need effective f_{ce} (including resistors) for (a)/(b)

where $|A_n(jf)|$ is constant (2.83 V/V)

$$\textcircled{a} \textcircled{b} e_{no}^2 \approx 2.83^2 \left(4 \times 10^{-13} / f + 2.05 \times 10^{-13} \right)$$

$$= 1.64 \times 10^{-12} (1.95/f + 1), \quad f_{ce} \approx 1.95$$

since $f_{ce} \ll 10 \text{ Hz} = f_L$ (lower integration limit)

then

$$e_{no}^2 \approx 1.64 \times 10^{-12} \text{ V}^2/\text{Hz}$$

→ $\textcircled{a}$ doesn't matter, therefore effectively white noise → use NEB!

$$\text{NEB} = f_B / 2$$

$$= 1.66 \text{ MHz}$$

since $f_L \ll f_B$, assume $f_L = 0$

$$E_{no}^2 = e_{no}^2 \text{NEB} = 2.73 \times 10^{-6}$$

$$E_{no} = 1.65 \text{ mV}$$

6050 →

② 6050 Values

$$f_B = 141 \text{ kHz}$$
$$e_n^2 = (20 \text{ nV}/\sqrt{\text{Hz}})^2 \left(\frac{100 \text{ Hz}}{f} + 1 \right)$$
$$i_{nn}^2 = i_{np}^2 = (0.05 \text{ pA}/\sqrt{\text{Hz}})^2$$
$$A_n(jf) = \frac{2.83}{1 + f/141 \text{ k}}$$
$$|A_n(jf)| = \frac{2.83}{\sqrt{1 + f^2/141 \text{ k}^2}}$$

$$e_{ni}^2 = e_n^2 + \left(\frac{R_{eq} R_f}{R_{eq} + R_f} \right)^2 [4kT + i_{nn}^2]$$
$$e_{ni}^2 = (4 \times 10^{-16}) \left(\frac{100}{f} + 1 \right) + (1.06 \times 10^{-7}) (1.645 \times 10^{-20} + 2.5 \times 10^{-27})$$

$\approx 1.645 \times 10^{-20}$

$$e_{ni}^2 = \frac{4 \times 10^{-14}}{f} + 1.75 \times 10^{-13}$$

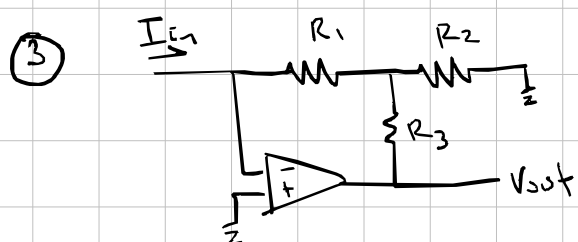
$$e_{no}^2 = \frac{2.83^2}{1 + f^2/141 \text{ k}^2} \left(\frac{4 \times 10^{-14}}{f} + 1.75 \times 10^{-13} \right)$$

$$E_{no}^2 = \int_{10}^{100 \text{ M}} e_{no}^2 df$$

Numerical Integration:

$$E_{no}^2 = 3.101 \times 10^{-7} \text{ V}^2$$

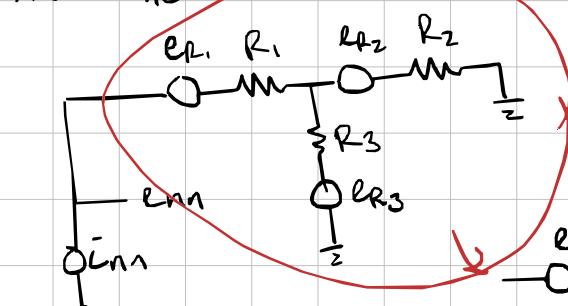
$$E_{no} = 0.56 \text{ mV}$$



from lecture 5:

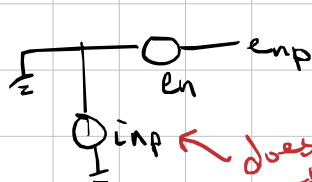
$$A = -R_1 \left(1 + \frac{R_3}{R_1} + \frac{R_3}{R_2} \right)$$

find e_{ni} :



$$R_{eq} = R_1 + R_2 \parallel R_3 = R_1 + \frac{R_2 R_3}{R_2 + R_3}$$

$$\rightarrow e_{Reg}^2 = 4kT R_{eq}$$



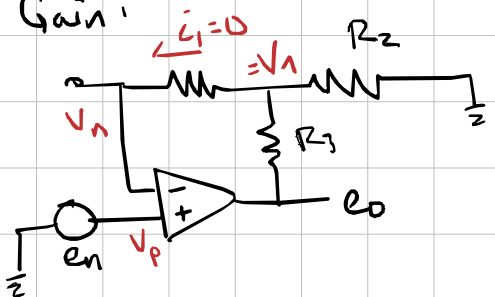
$$e_{ni}^2 = e_{Reg}^2 + i_{nn}^2 R_{eq}^2$$

$$e_{np}^2 = e_n^2$$

$$e_{ni}^2 = e_{Reg}^2 + i_{nn}^2 R_{eq}^2 + e_n^2$$

$$e_{no}^2 = |A_n(jf)|^2 e_{ni}^2$$

Noise Gain:



$$V_n = V_p$$

$$V_n = \frac{R_2}{R_2 + R_3} e_i$$

$$\beta_n = R_2 / (R_2 + R_3) = 1/A_n$$

$$A_n = 1 + \frac{R_3}{R_2}, A_n(jf) = A_n / (1 + jf/f_{Bn})$$

from GBWP: $f_{Bn} = f_t \beta_n$

Numbers: $R_1 = 20k, R_2 = 1k, R_3 = 15k, R_{eq} \approx 21k$

$$A = -335000 V/A = -335 V/mA$$

$$\beta_n = 0.0625, f_{Bn} = 0.0625 f_t, A_n = 16$$

4050

$$f_{Bn} = 0.0625 \cdot 3MHz = 187.5kHz$$

$$i_{nw} = 0$$

$$E_o^2 = \int_{0.1Hz}^{\infty} \left(16^2 / \left(1 + \frac{f^2}{187.5k^2} \right) \right) \left(e_{Reg}^2 + e_{nw}^2 \left(\frac{f_{ce}}{f} + 1 \right) \right) df$$

$$\text{into matlab} \rightarrow E_o^2 = 1.29 \times 10^{-9} V^2 \rightarrow E_o = 36 \mu V$$

6050

$$f_{Bn} = 0.0625 \cdot 400k = 25kHz$$

$$i_{nw} = 0.05 pA/\sqrt{Hz}$$

$$E_o^2 = \int_{0.1Hz}^{\infty} \left(16^2 / \left(1 + \frac{f^2}{25k^2} \right) \right) \left(e_{Reg}^2 + i_{nw}^2 R_{eq}^2 + e_{nw}^2 \left(\frac{f_{ce}}{f} + 1 \right) \right) df$$

$$\text{into Matlab} \rightarrow E_o^2 = 1.23 \times 10^{-10} \rightarrow E_o = 11.1 \mu V$$

SNR: Need V_{rms} of input for $\sin$, square, tri:

$$SNR = 20 \log_{10} \left(\frac{V_{SRMS}}{E_o} \right)$$

$\frac{1}{\sqrt{2}}$ $\frac{1}{1}$ $\frac{1}{\sqrt{3}}$
wikipedia

Assume $I_{in} < f_B$, then

$$V_o \approx A I_{in} = (-335000)(10\mu) = 3.35V$$

(peak amplitude)

-4050

	sin	square	tri
$V_{p,rms}$	2.37V	3.35V	1.93V
SNR	96.4dB	99.4dB	94.6dB

-6050

	sin	square	tri
$V_{p,rms}$	2.37V	3.35V	1.93V
SNR	106.6dB	109.6dB	104.8dB